

Atrial Fibrillation & Stroke Prevention

Risk Stratification, Prevention Strategies, Emerging Evidence, & Knowledge Gaps
(Neurologist Perspective)

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Objectives

1

Evaluate stroke risk stratification tools including performance metrics & limitations

2

Understand stroke prevention strategies: anticoagulation, timing after stroke, alternatives

3

Introduce emerging approaches and select knowledge gaps in AF-related stroke prevention

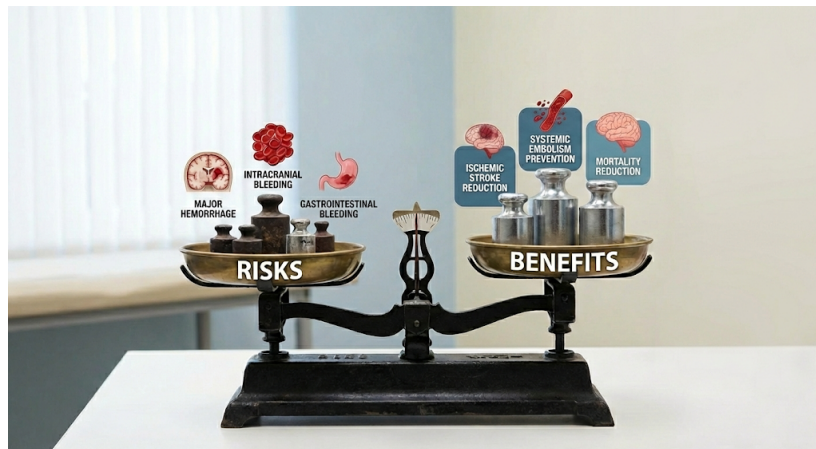
Clinical Risk Stratification

AF & Stroke Risk

AF ↑ stroke risk ~5X

- Absolute risk varies widely

Anticoagulation ↓ stroke risk ~60-70%, but ↑ bleeding



Decision Framework & Treatment Threshold

Net Clinical Benefit = stroke risk reduction – major hemorrhage risk increase

CHA₂DS₂-VASc Risk Score

TABLE 4 CHA₂DS₂-VASc Score for Prediction of Stroke in Atrial Fibrillation Patients

C	Congestive heart failure	1
H	Hypertension, including treated / controlled	1
A	Age ≥75 yrs	2
D	Diabetes mellitus	1
S ₂	Prior TIA or stroke	2
V	Vascular disease (MI, aortic plaque, and so on)	1
A	Age 65-74 yrs	1
Sc	Sex category (female = 1 point)	1

MI = myocardial infarction; TIA = transient ischemic attack.

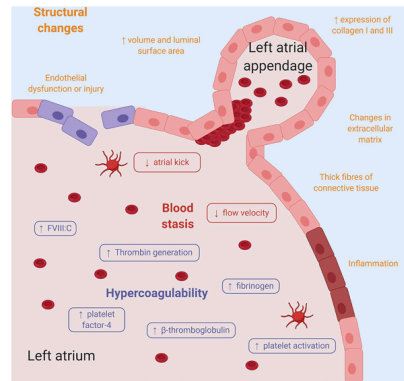


Figure 1 Effects of atrial fibrillation on the various elements of Virchow's triad (hypercoagulability, structural changes and blood stasis).

Calibration matters

Absolute risk depends on cohort and endpoint.

The original annual-risk bars were not traced to a primary cohort and are withheld as individual risk estimates.

See Repair 15.

J Am Coll Cardiol. 2020 Apr 14;75(14):1689-1713.

Eur Heart J. 2012 Jun;33(12):1500-1510.

Circulation. 2023 Dec 5;148(23):e1-e125.

Heart. 2020 Oct;106(19):1463-1468.

CHA₂DS₂-VASc: Performance Characteristics

Discrimination & Calibration

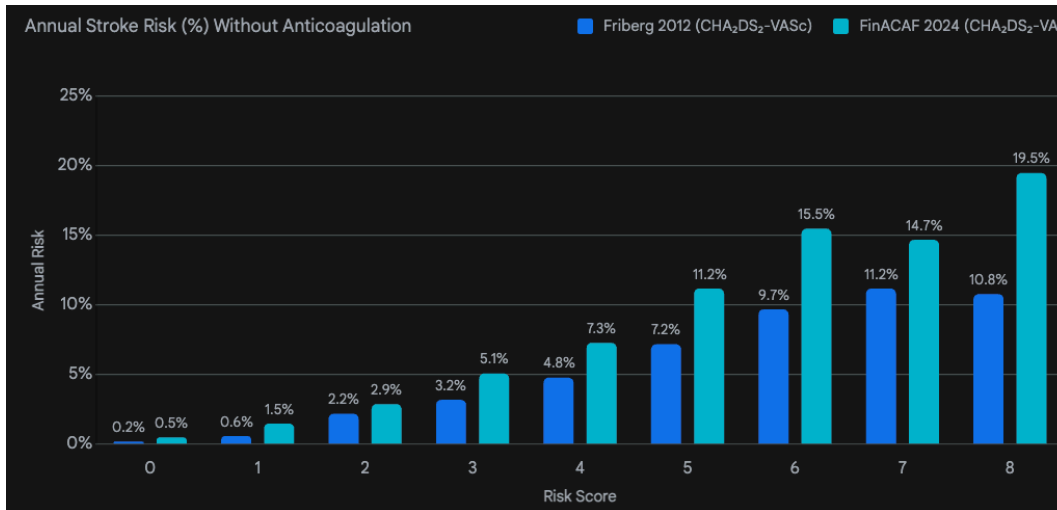
- C-statistic: ~0.65 (modest)
- Variable across populations
- Overestimates in certain subgroups

Limitations

- Binary variables (loss of granularity)
- No AF burden/type or renal function

Why is it clinically useful?

- High negative predictive value (identifies true “low risk” from “not low risk”) → is risk low enough to avoid anticoagulation?
- Simplicity → clinical factors, reproducible, scalable



FinACAF: CHA₂DS₂-VA, 2017-2018; Friberg: CHA₂DS₂-VASc. Different scores/cohorts: not a temporal trend. Repair 19.

2024 ESC: CHA₂DS₂-VA

- Female sex is modifier (not independent risk factor)

Circulation. 2012 May 15;125(19):2298-307.
Lancet Reg Health Eur. 2024 Jun 10;43:100967.

Bleeding Risk

HAS-BLED Score

Score ≥ 3 = High hemorrhage risk

C-index ~ 0.6

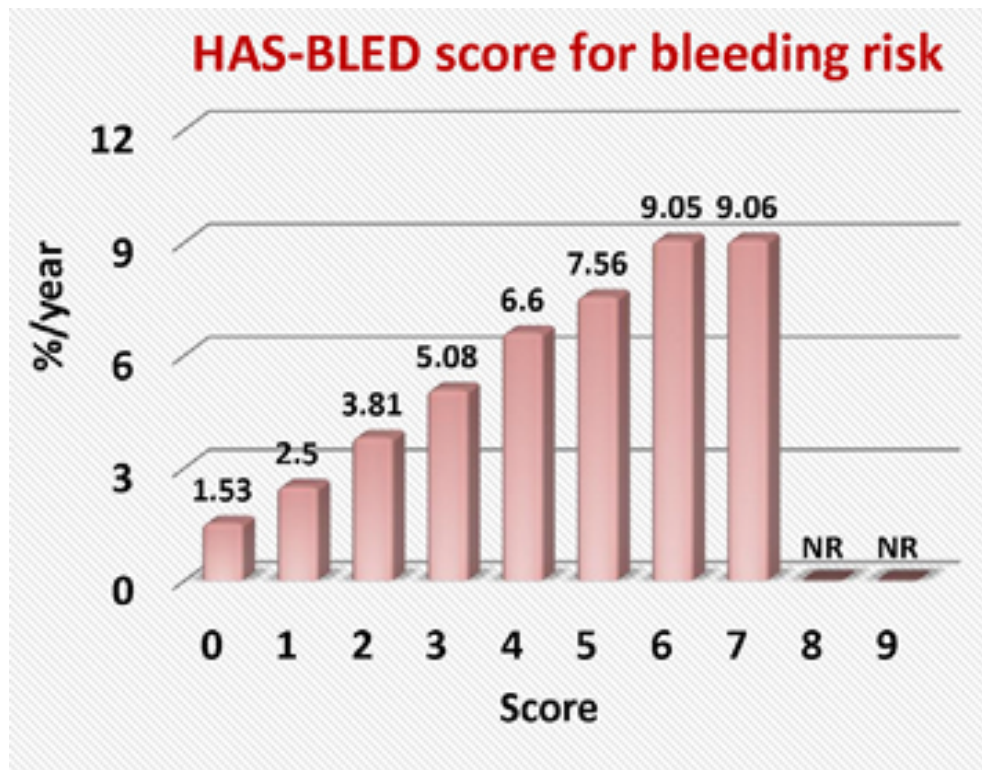
High hemorrhage risk $\neq$ "Don't anticoagulate"

- Identify & address modifiable risk factors
- Guide monitoring intensity

Often hemorrhage risk < stroke risk

* Developed in warfarin treatment era

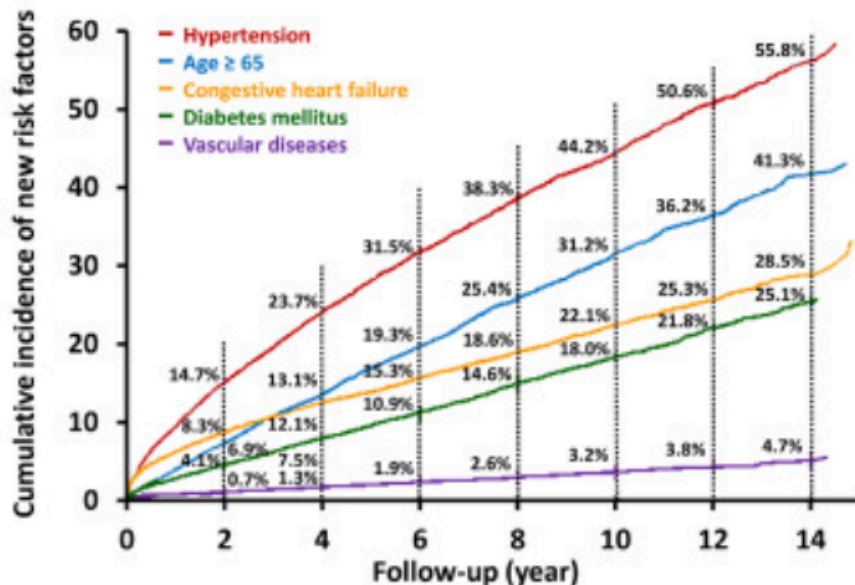
Condition	Points
H – Hypertension	1
A – Ab(N) liver/renal	1 point each
S – Stroke	1
B – Bleeding	1
L – Labile INRs	1
E – Elderly (>65)	1
D – Drugs or ETOH	1 point each



Am J Med. 2020 Oct;133(10):1195-1202.e2.
Lancet Reg Health Eur. 2024 Feb 1;37:100797.
Chest. 2010 Nov;138(5):1093-1100.
Eur Heart J. 2012 Jun;33(12):1500-1510.

Risk is Dynamic

Cumulative incidence of new stroke risk factors for patients with a baseline CHA₂DS₂-VASc score of 0 (M) or 1 (F)



Multimorbidity → Δ net clinical benefit of treatment

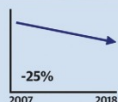
- Age
- Accumulation risk factors for stroke & hemorrhage

CENTRAL ILLUSTRATION: Decreasing Nonanticoagulated Stroke Risk in Atrial Fibrillation Patients Despite Rising Age and Stroke Risk Scores

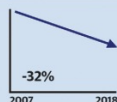


- Nationwide cohort study covering all levels of care in Finland
- Observation period from 2007 to 2018
- Mean CHA₂DS₂-VA score increased from 2.5 to 3.0 ↑
- Mean age increased from 70 to 73 years ↑
- Nonanticoagulated stroke risk decreased by 25% ↓

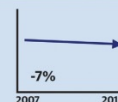
All Patients



Women



Men

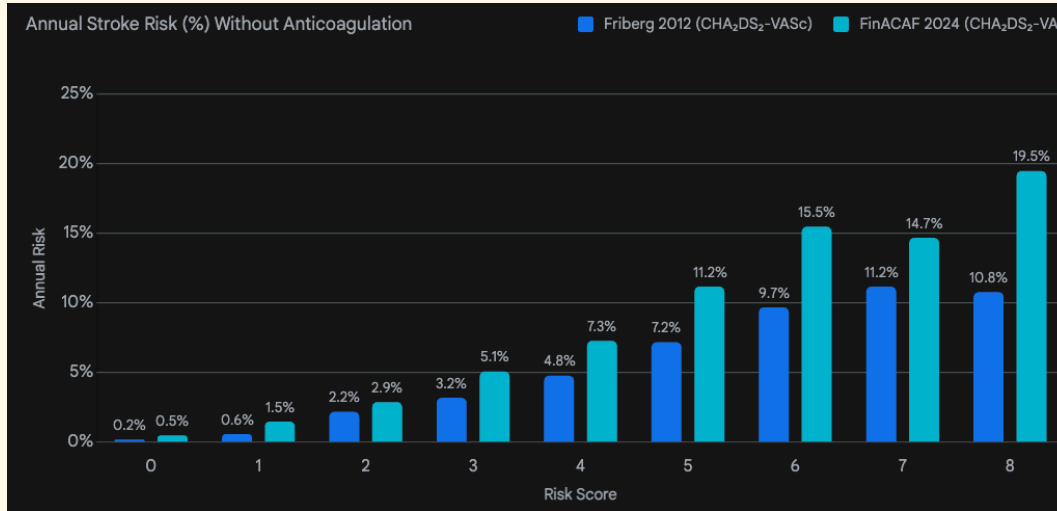


Teppo K, et al. JACC Clin Electrophysiol. 2025;11(3):583-592.

J Am Coll Cardiol. 2018;71(2):122-132.
Thromb Haemost. 2018 Jul;118(7):1296-1304.
Eur Heart J. 2024 May 27;45(20):1819-1827.



Discussion Question



Discussion

How do different score definitions, cohorts and competing-risk methods limit this comparison?

Teal: FinACAF CHA₂DS₂-VA, 2017-2018 stratum.

Blue: Friberg CHA₂DS₂-VASc.

Publication years are not observation years. The contrast cannot be attributed to time alone.

Anticoagulation

DOACs vs. Warfarin

- DOACs > warfarin
- Among DOACs → no direct comparative trials
- Apixaban preferred (↓ hemorrhage risk)

Warfarin: mechanical valves; rheumatic MS or MS of moderate or greater severity.

* Aspirin alone NOT recommended

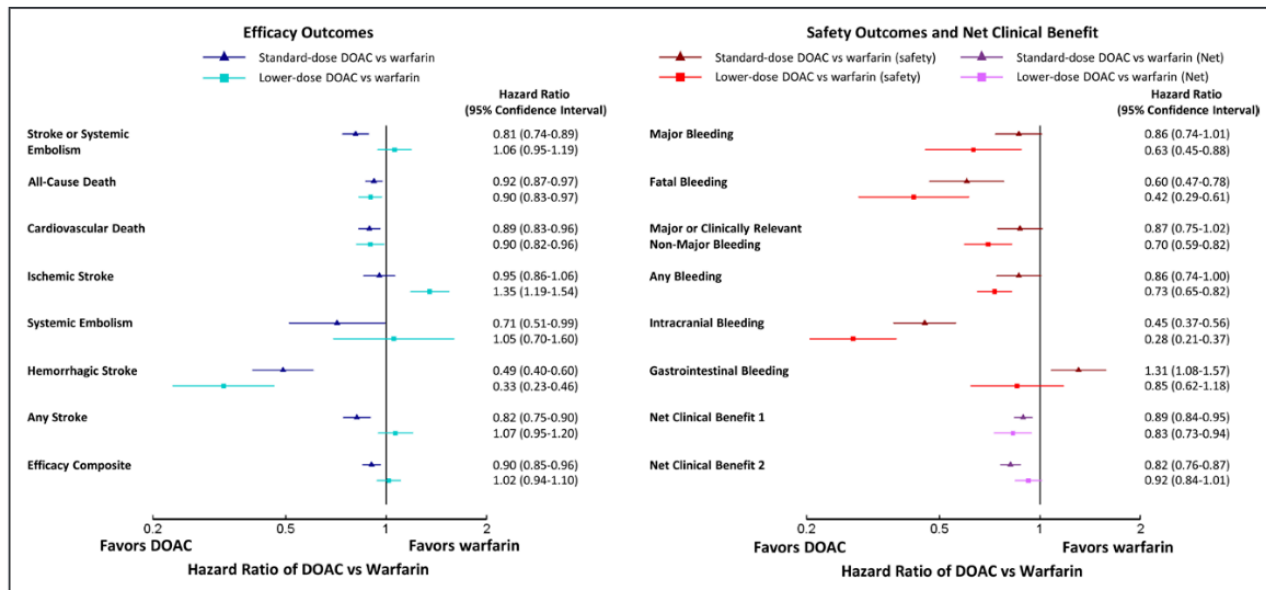


Figure 1. Comparison of standard-dose and lower-dose direct oral anticoagulants vs warfarin for efficacy, safety, and net clinical benefit outcomes.

Comprehensive Atrial Fibrillation Management

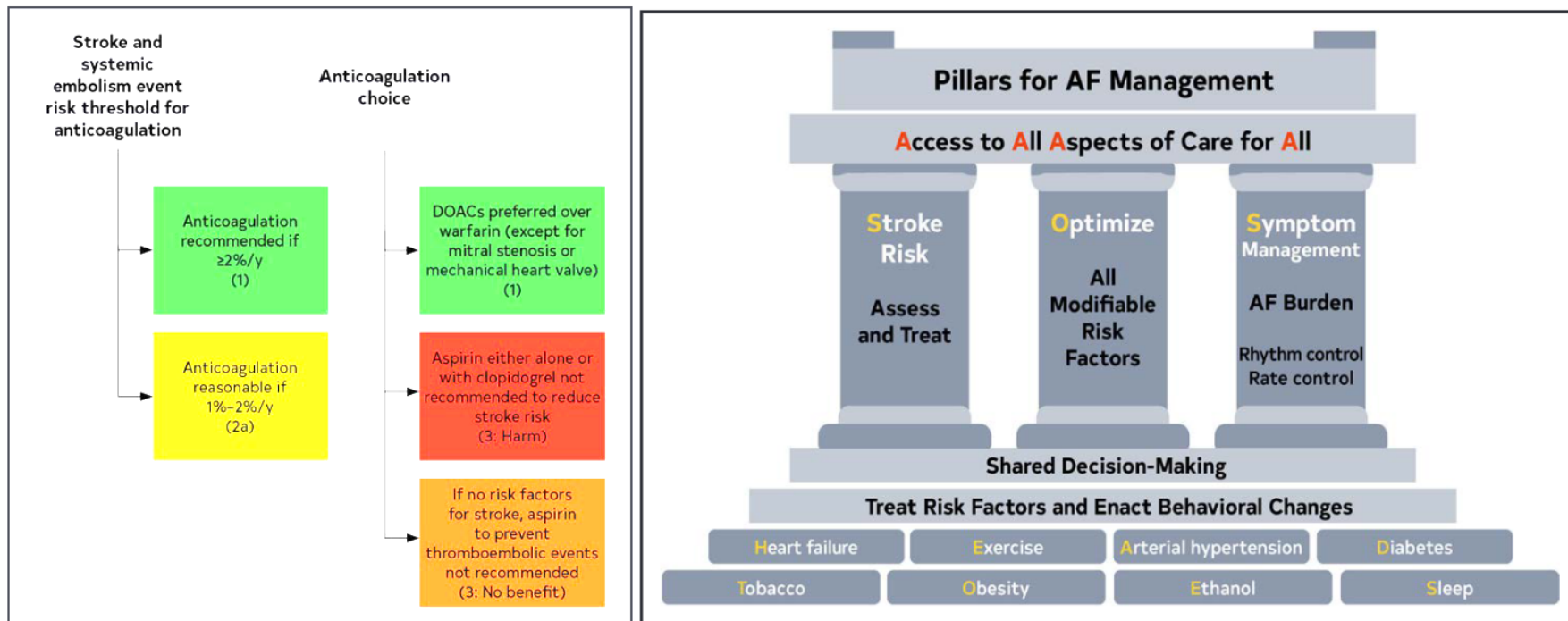


Figure 10. Antithrombotic Options in Patients With AF.

Timing After Acute Ischemic Stroke

CATALYST IPD meta-analysis (n=5,441)

Individual participant data meta-analysis of 4 trials
(TIMING, ELAN, OPTIMAS, START)

- Population: acute ischemic stroke + AFib
- Treatment/Comparator: DOAC initiation ≤ 4 days (early) or ≥ 5 days (late)
- Outcome: (composite) recurrent ischemic stroke or symptomatic ICH in ≤ 30 days

→ Early DOAC initiation reduced risk

- Limited generalizability to severe stroke, hemorrhagic transformation

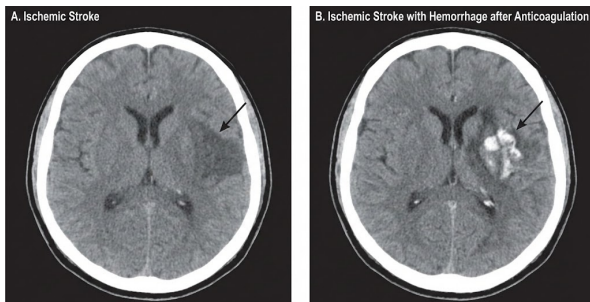


Figure 1. Comparison of head CTs: (A) Acute ischemic stroke showing hypodensity. (B) Hemorrhagic transformation within the ischemic region. Intracranial hemorrhage is indicated by the hyperdense blood.

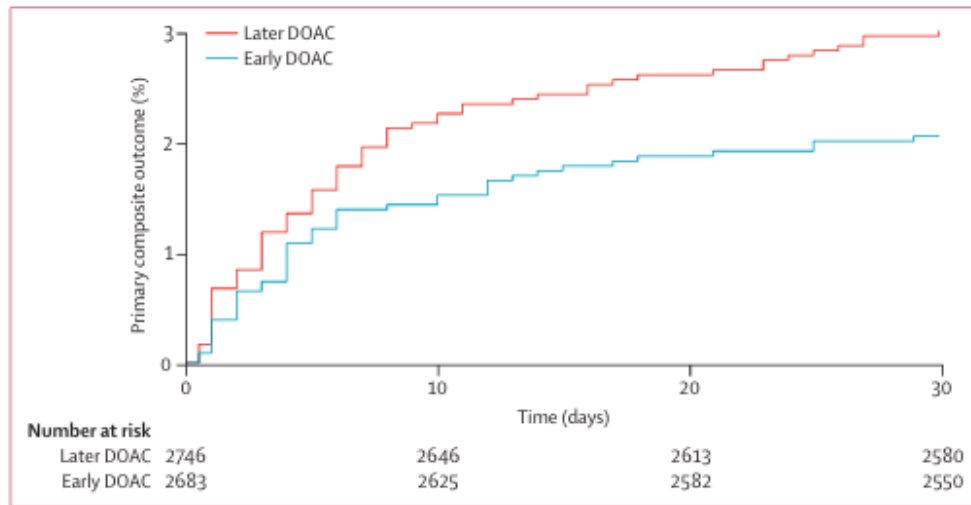


Figure 2: Primary composite outcome at 30 days by DOAC initiation timing (cumulative hazard)

Log-rank p value: 0.034. DOAC=direct oral anticoagulant.

- Small-moderate infarct → $\leq 2-4$ days
- Large infarct → 1-2 weeks

Lancet. 2025 Jul 5;406(10498):43-51.

Lancet Neurol. 2024 Apr;23(4):404-417.

Image generated with Google Gemini

Low-Burden AF

- Asymptomatic
- Device-detected
- 6 minutes – 24 hours
- Lower stroke risk than clinical AF
- AC decisions require more nuanced shared decision-making

Use the specified trial endpoint

AVERROES - clinical AF unsuitable for warfarin

Aspirin / apixaban, % per year:

Stroke or systemic embolism: 3.7 / 1.6.

Major bleeding: 1.2 / 1.4.

ARTESIA - device-detected subclinical AF

Aspirin / apixaban, % per patient-year:

Stroke or systemic embolism: 1.24 / 0.78 (intention-to-treat).

Major bleeding: 0.94 / 1.71 (on-treatment).

Different populations and analysis sets: do not rank treatments across trials or use time-unspecified NNT/NNH. See Repair 11.

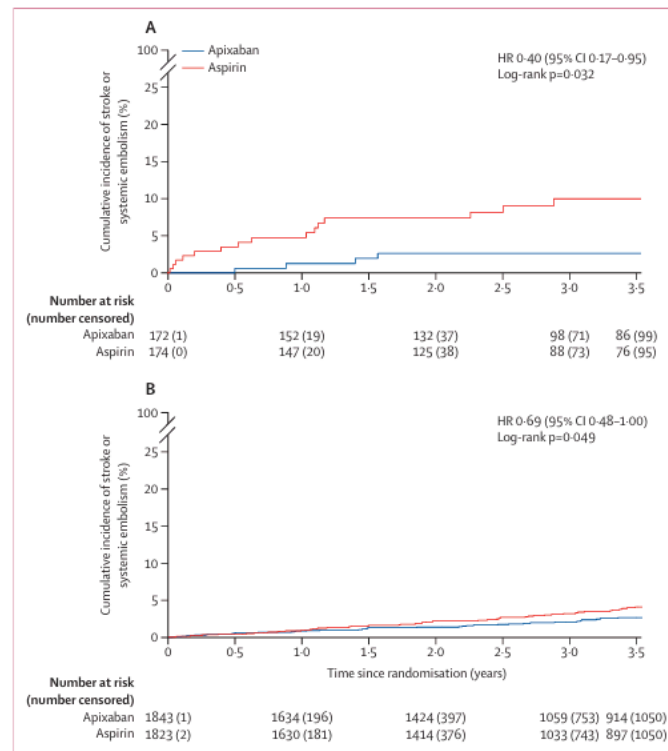


Figure 2: Time to stroke or systemic embolism during 3.5 years of follow-up in participants with a history of stroke or transient ischaemic attack (A) and those without (B) according to treatment assignment (apixaban vs aspirin)

HR=hazard ratio.

Left Atrial Appendage Occlusion (LAAO) - Percutaneous

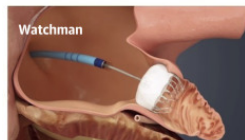
Rationale

~80-90% of AF thrombi form in the LAA

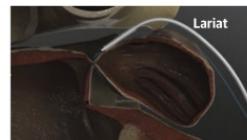
Occlude LAA → eliminate source of most thromboembolism → Avoid long-term anticoagulation

- Considered for ABSOLUTE contraindication to long-term AC → NOT first-line alternative to DOACs
- Non-inferiority of LAAO to DOACs uncertain (high-risk groups)
- Requires post-procedure anticoagulation (or dual AP treatment)

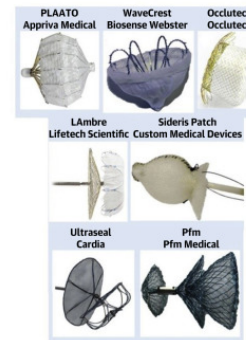
Endocardial LAA Occluders



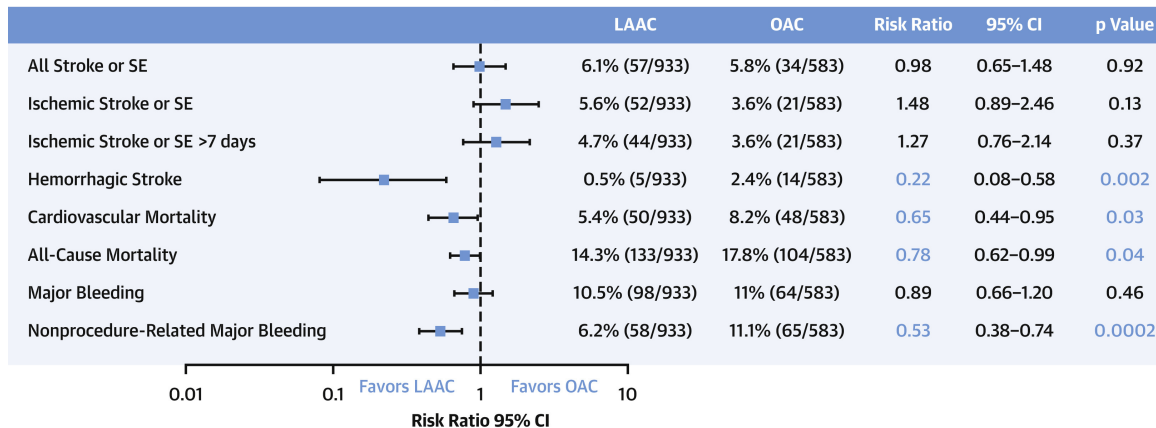
Epicardial LAA Excluders



Other Endocardial LAA Occluders



Summary Plot of Clinical Outcomes





Discussion Question

Left Atrial Appendage Occlusion (LAAO) - Surgical

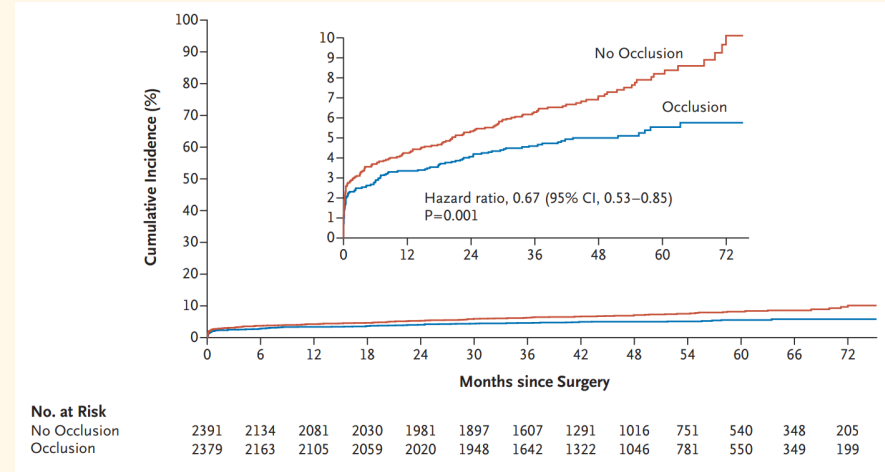
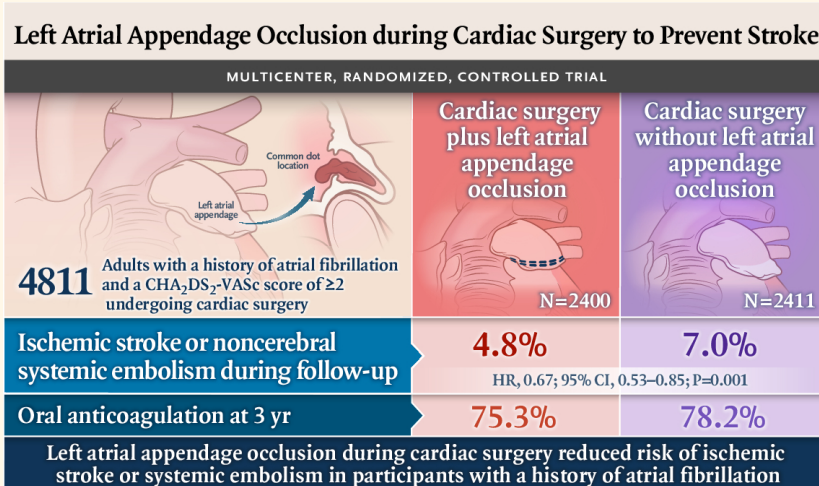


Figure 1. Cumulative Incidence of Stroke or Systemic Arterial Embolism.

Mean CHADS₂VASc ~4

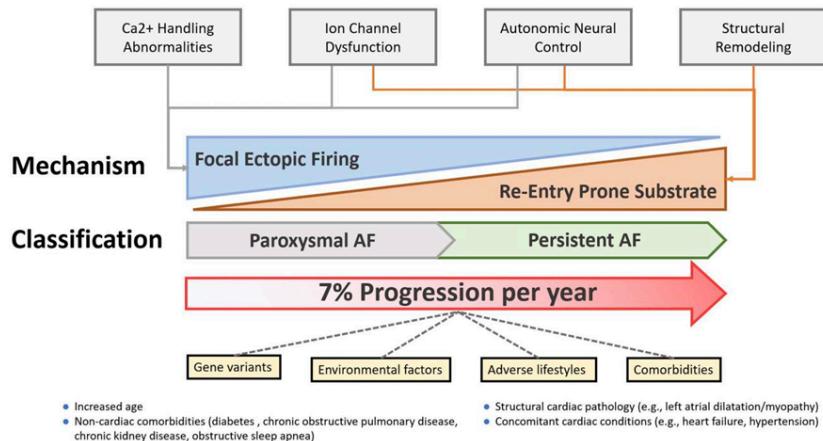
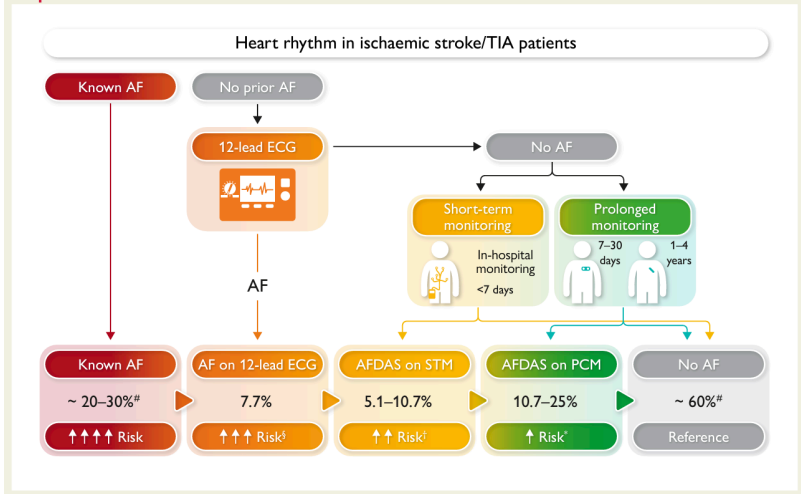
~77% on AC @ 3 y

Why was AC continued in most?
How should findings be interpreted?

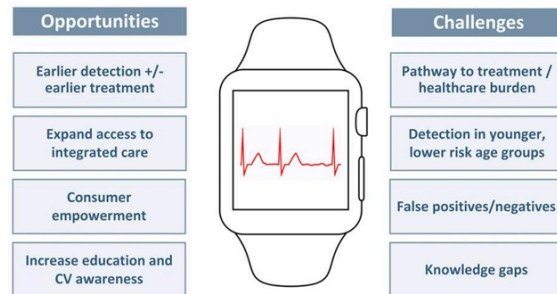
- Lack of clinical equipoise
 - Real-world uncertainty
 - How were 23% that stopped AC different?
- Supports LAAO + AC may further reduce stroke risk

Knowledge Gap: Detection – Outcome Disconnect

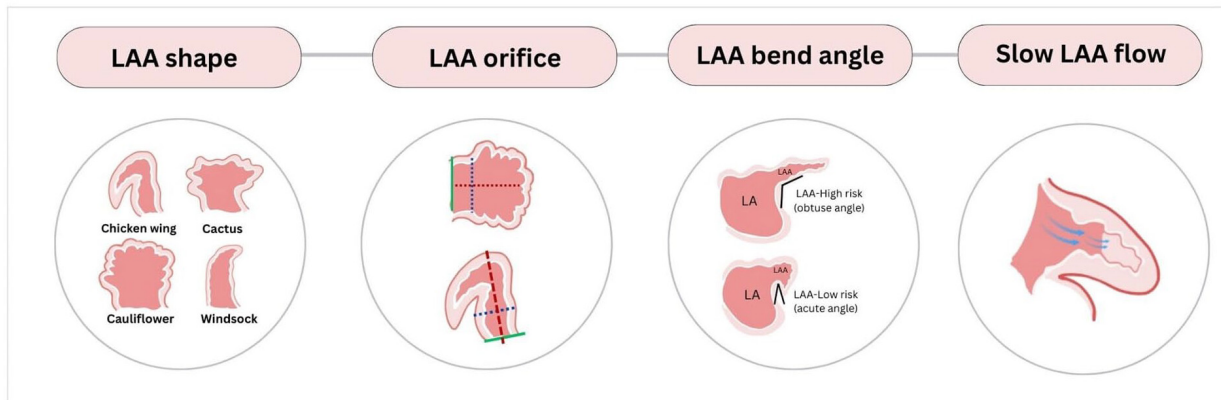
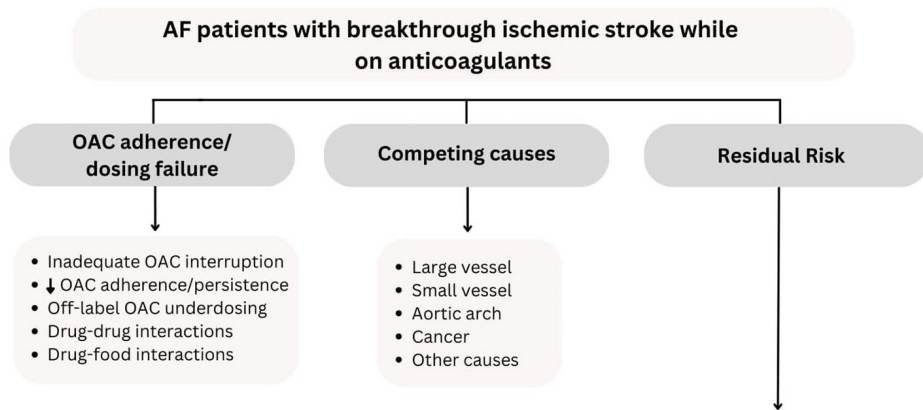
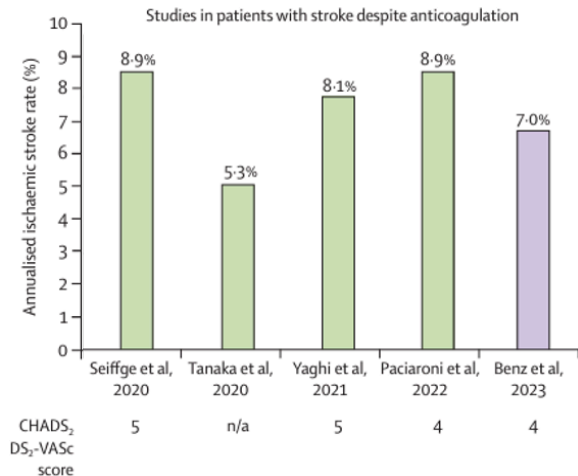
Graphical Abstract



- Screening ↑ detection, BUT early detection ≠ ↓ stroke/mortality
- Low-burden → higher burden
- Treatment threshold uncertain



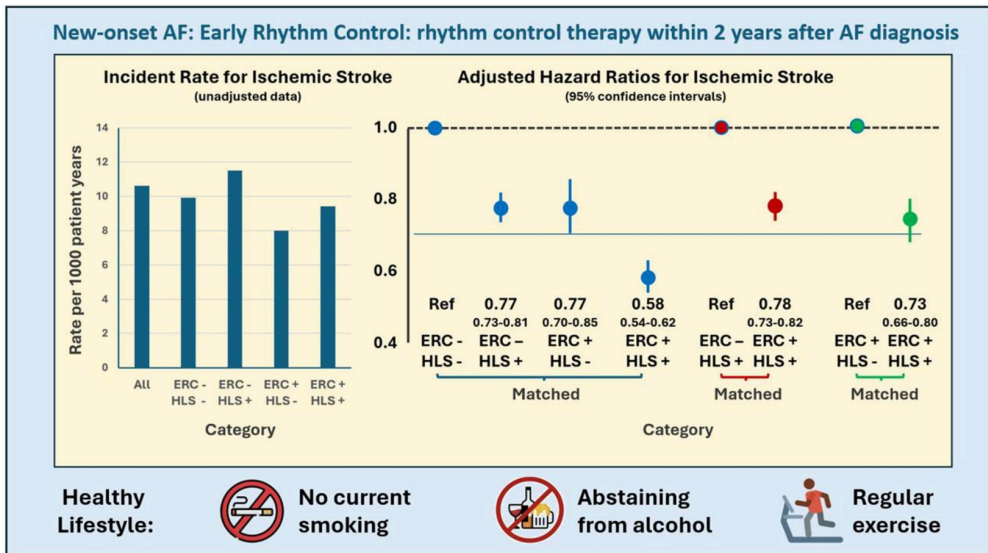
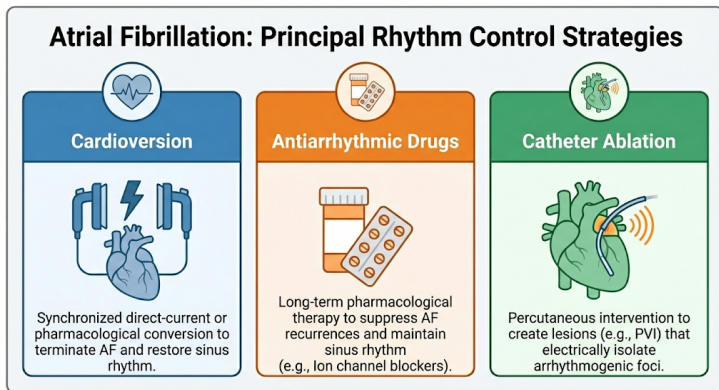
Knowledge Gap: Recurrent Stroke Despite AC



🛑 Switching DOACs or
addition of aspirin 🖐️

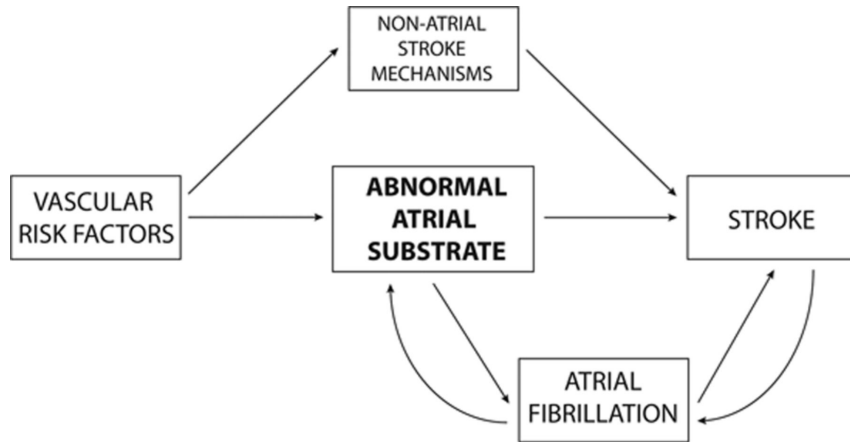
- LAAO + AC
- Factor XI inhibitors

Knowledge Gap: Early Rhythm Control

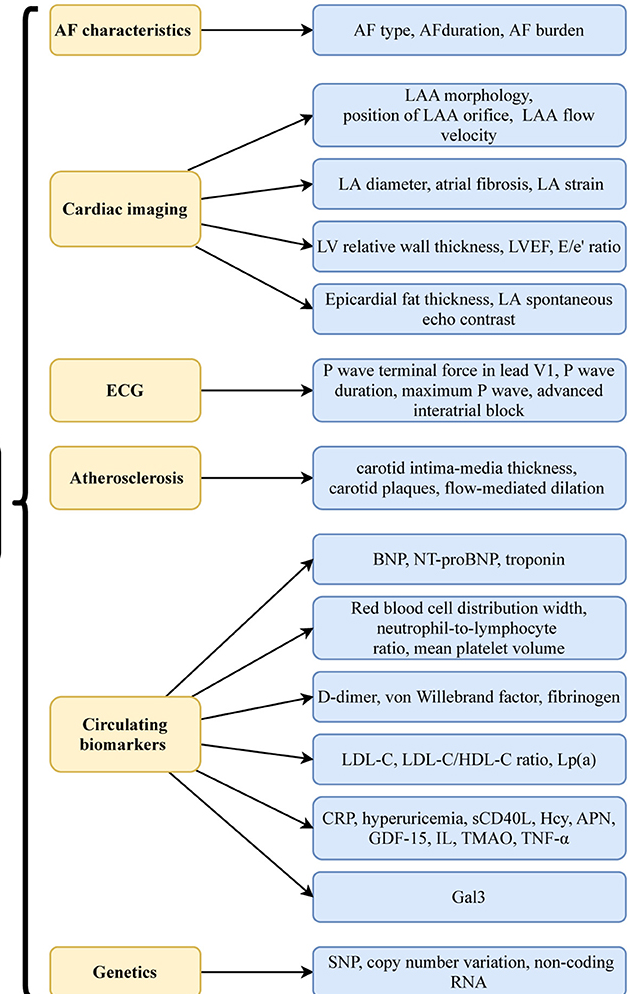
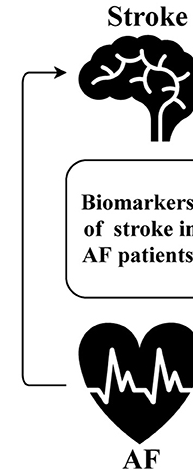


- ↓ stroke risk (by ↓ AF burden)
- Successful restoration of sinus rhythm ≠ discontinuation of AC
 - High risk of recurrence
 - Subgroup? – early onset, low-burden, low CHA₂DS₂VASC

Knowledge Gap: Risk (& Therapeutic) Biomarkers



- Cardiac Structural
- Electrophysiologic (EKG markers)
- Circulating (proteins, metabolites)



Questions?



Targeted repairs | 30 September 2026

The original deck is an archive. These numbered notes document targeted corrections; unchanged material has not been comprehensively updated. Trial selection, source conflicts and unread correction notices remain explicit. Original slide numbers are preserved.

Repair 11 - Cross-trial comparison: endpoints and analysis populations

Keep endpoints and analysis populations separate. AVERROES: stroke/systemic embolism was 3.7%/year with aspirin and 1.6%/year with apixaban; major bleeding was 1.2%/year and 1.4%/year, respectively. ARTESIA: primary stroke/systemic embolism was 1.24%/patient-year with aspirin and 0.78%/patient-year with apixaban (intention-to-treat); major bleeding was 0.94% and 1.71%/patient-year (on-treatment). The 1.02%/0.64% pair in ARTESIA was ischemic or unknown stroke, not its primary endpoint. The original table combined different populations/endpoints and time-unspecified NNT/NNH; it has been replaced without cross-trial efficacy ranking.

AVERROES | <https://pubmed.ncbi.nlm.nih.gov/21309657/>

ARTESIA | <https://pubmed.ncbi.nlm.nih.gov/37952132/>

ARTESIAfull | <https://openaccess.sgul.ac.uk/id/eprint/115973/1/nejmoa2310234.pdf>

Repair 13 - Valve disease: warfarin exception scope

The AF2023 warfarin exception specifies rheumatic mitral stenosis or mitral stenosis of moderate or greater severity, and mechanical valves. It does not apply to all rheumatic valve disease or all degrees of nonrheumatic mitral stenosis. Source: section 6.8.5, recommendations for AF in valvular heart disease.

AF2023 | <https://pmc.ncbi.nlm.nih.gov/articles/PMC11095842/>

Repair 15 - Stroke-risk scores: hypertension and calibration

CHA2DS2-VASc hypertension includes current antihypertensive treatment, including controlled treated hypertension; it is not limited to a currently elevated blood pressure. Source: AF2023 Table 10 risk-factor definition. Original absolute-risk bar values were not traced to a primary cohort; they have been withheld as individual risk estimates. Absolute risk varies with population, endpoint, follow-up and competing risks.

AF2023 | <https://pmc.ncbi.nlm.nih.gov/articles/PMC11095842/>

Repair 19 - Risk-score comparisons: cohort and method differences

The teal bars reproduce FinACAF CHA2DS2-VA estimates for the 2017-2018 stratum; the blue comparator is Friberg CHA2DS2-VASc. These are different score definitions and cohorts, with different methods and competing risks. Publication years (2024 and 2012) are not enrollment years. The contrast cannot be attributed to time alone. The teaching question now asks how these design differences limit comparison; proposed causal explanations are not established findings.

FinACAF | <https://pmc.ncbi.nlm.nih.gov/articles/PMC11337097/>

Friberg | <https://pubmed.ncbi.nlm.nih.gov/22514252/>

Repair 20 - CATALYST: pooled sample identity

The figure and associated pooled analysis are CATALYST, an individual participant data meta-analysis of 5,441 participants. The hidden ELAN Trial (n=2,013) heading and overlaid text have been removed from the PDF content, not simply covered.

CATALYST | <https://pubmed.ncbi.nlm.nih.gov/40570866/>

ELAN | <https://pubmed.ncbi.nlm.nih.gov/37222476/>